Branson Tay

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WORK EXPERIENCE

Jaguar TCS Racing **Aug. 2024 – Aug. 2025**
Junior Strategy & Software Engineer *Kidlington, United Kingdom*
Stack: Python, MATLAB, ReactJS, Typescript, CSS, MongoDB, Azure Pipelines, Docker, VBA

- Develop full-stack ReactJS apps visualising data critical to race strategy using live telemetry, tyre & timing data.
- Built and deployed a Python data pipeline converting Timing Feed data in NoSQL to processed ATLAS telemetry channels, following Agile practices, for performance analysis, tow analysis, and competitor analysis.
    - Implemented versioning, deployment pipeline and sustainable integration into existing systems.
- Deployed dockerised trackside tools via Azure DevOps with robust CI/CD, testing, and error handling.
- Wrote a custom Python library to extract and write car data to ATLAS format using MAT Atlas DLLs letting the Performance group engage in ML/AI projects such as virtual sensors, or large aggregations of car data.
- Troubleshoot and continuously improve existing codebases and race-critical tools live during race-weekends.
- Process pre-race battery simulations and simulator data results to monitor battery temperatures and ensure legality, correlating with track data to ensure accuracy in our models and to predict race temperature profiles.
- Logged and streamed live telemetry from WinTax using Visual Basic Scripts to a TCP socket via REST APIs.
- Perform qualitative competitor radio analysis (energy management, pit-stops, strategy, battery temperature management, vehicle setup and tyre run-plans) delivering concise per-session summaries to strategists.
- Analysed driver telemetry in ATLAS to optimise energy management and performance to gain race pace.

Republic of Singapore Navy **Nov. 2020 – Jul. 2022**
Marine Systems Specialist Technician *Singapore, Singapore*

- Trained on Singapore Navy's largest marine vessel, on maintenance and wartime awareness of ship systems.
- Led cross-team war-time firefighting simulation aboard ship engine operations room to assign appropriate fixes.
    - Analysed live sensor data in core ship systems from telemetry displays under high-pressure situations.
- Contributed to 50 pre- and post-mission debriefs, highlighting potential improvements for team efficiency.

Travelindr **Oct. 2019 – Mar. 2020**
Co-founder & CEO *Singapore, Singapore*

- Founded Travelindr, a group travel-itinerary generator, with a small team achieving a Minimal-Viable Product.
- Oversaw core strategies, product roadmap, data & competitor analysis, talent acquisition & market research.
- Conduct focus groups for qualitative data, pitched to investors, incubated in esteemed multi-national company.

EDUCATION

University of Leeds **Sep 2022. – Jul. 2026**
BSc Computer Science with Year in Industry
Leeds, UK

- **Formula Student Performance & Simulation Sub-team Lead**: Lead a diverse team of 12, with the Software Development Lifecycle and Agile, in projects like thermal management modelling, lift & coast simulators to quasi-static lap time simulators, ensuring the team follows a data-driven approach for design considerations.
- **Proposed Final Year Project**: Building a race strategy simulation platform using multi-agent reinforcement learning for the F1 2026 regulations. Models key race dynamics including tyre wear, overtaking difficulty, and energy deployment, investigating different algorithms and state spaces for optimal policy convergence.

Ngee Ann Polytechnic **Jan. 2017 - Mar. 2020**
Diploma in Mechanical Engineering *Singapore, Singapore*

- Utilised AutoCAD CAD software to fabricate gearbox given specific load, speed, and dimensional requirements
- Recognized with the Good Progress Award and the School of Engineering Merit Award.

TECHNICAL PROJECTS

Formula Student Lap Time Simulator | Python, Git, Vehicle Dynamics, Optimisation

- Developed quasi-static steady-state lap time simulators with Object-Oriented Programming using TTC tyre data, enabling sensitivity analysis of design parameters such as track width, wheelbase, and chassis stiffness.

2D CFD Formula Student Radiator Optimisation Simulator | Rust, Aerodynamics, Numerical Methods

- Developed a 2D incompressible fluid solver analysing radiator angle effects on cooling, pressure drop and drag.
- Automated batch simulations and visualisations to optimise radiator design, leveraging parallel processing.

Formula Student ICE Data Acquisition System | C++, Git, Arduino

- Aggregated data from GPS module to Arduino Teensy, visualising live data to the driver dash and over radio frequency for multiple departments to better quantify and enhance vehicle performance.
- Honed proficiency in collaborative software engineering tools like Git and conducting code reviews.

Formula Student EV Battery Management System | C++, CAN Bus protocol

- Spearheaded EV emulation of Battery Management System temperature sensor extension embedded system.
- Architected an expansion module using the CAN bus protocol for robust and efficient data transfer.
- Designed microprocessor framework to process battery module voltages to tangible data in C++.

Pathfinder Visualiser | ReactJS, Flask, HTML, CSS, JS

- Programmed a Full-Stack Web Application with Flask serving as a REST API and React as the frontend.
- Employed Dijkstra's algorithm and A* algorithm to visually demonstrate the speed difference between them.

ADDITIONAL EXPERIENCE

E-commerce Shopify Store Building | Facebook Ads Manager, Adobe Photoshop, Adobe Premiere Pro

- Developed and marketed multiple e-commerce shop brands by creating catchy, attention-grabbing creatives.
- Drove user attention by utilising good design principles like obvious action buttons, and consistent elements.
    - Increased conversions in by 10% by adapting "add-to-cart" button colour based on cursor heatmaps.
- Constructed efficient sales funnel, extracting data insights, and targeting trends, fully utilising data analysis tools.

Startup Partnership Intern, Global Entrepreneurial Internship Scholar

- Pioneered Voucher System project integrating cross-department efforts within a strict one-month window.
    - Achieved a 10% surge in the 30,000 annual customer base nationally.
- Independently secured over 10 successful strategic partnerships through presentations with senior executives.

TECHNICAL SKILLS, SOFT SKILLS & CERTIFICATIONS

Languages: Python, Matlab, HTML, CSS, Typescript, VBS, MongoQL, Rust, Java, C++, C.
Frameworks and Libraries: NumPy, SciPy, Bootstrap, Matplotlib, ReactJS, NodeJS, MongoDB, MySQL, Flask.
Tools: Docker, Git, Git Bash, Copilot, Azure DevOps, Azure Pipelines, Visual Studio, Microsoft Suite, ATLAS.

Soft Skills: Collaborative | Time-management | Curious | Adaptable | Multiperspectivity

Certifications & Awards

- Completed Harvard's introductory course to computer science, CS50.
- Runner-up in TRIVE Global Entrepreneurial Internship Program Startup Challenge.

HOBBIES & PERSONAL INTERESTS

- Engineering: Avid viewer of motorsport and airplane engineering videos | Karting.
- Stress-relieving: Padel | Table-tennis | Badminton | Cooking | Painting | Running.
- Enriching: Reading fiction & non-fiction books | Podcasts & videos on behavioural science.